
Uni Programming Languages Notes

Felicitas Pojtinger (fp036)

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1 Introduction

1.1 Contributing

These study materials are heavily based on [professor Ihler's "Aktuelle Programmiersprachen" lecture at HdM Stuttgart](#).

Found an error or have a suggestion? Please open an issue on GitHub (github.com/pojntfx/uni-programminglanguages-notes):



Figure 1: QR code to source repository

If you like the study materials, a GitHub star is always appreciated :)

1.2 License



Figure 2: AGPL-3.0 license badge

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2 Overview

2.1 General Design

- “A dynamic, open source programming language with a focus on simplicity and productivity. It has an elegant syntax that is natural to read and easy to write.”
- Inspired by Perl, Smalltalk, Eiffel, Ada, Lisp
- Multi-paradigm from the beginning: Functional, imperative and object-oriented
- Radical object orientation: Everything is an object, there are no primitive types like in Java (5 . times { print "We *love* Ruby -- it's outrageous!"})
- Very flexible, i.e. operators can be redefined
- Built-in blocks (closures) from the start, excellent mapreduce capabilities
- Prefers mixins over inheritance
- Syntax uses limited punctuation with some notable exceptions (instance variables with @, globals with \$ etc.)

2.2 Implementation Details

- Exception handling similar to Java & Python, but no checked exceptions
- Garbage collection without reference counts
- Simple C/C++ extension interface
- OS independent threading & Fibers, even if OS is single-threaded (like MS-DOS)
- Cross-platform: Linux, macOS, Windows, FreeBSD etc.
- Many implementation (MRI/CRuby, JRuby for Ruby in the JVM, TruffleRuby on GraalVM, mruby for embedded uses, Artichoke for WebAssembly and Rust)

2.3 Users

- Twitter
- Mastodon
- GitHub
- Airbnb
- Shopify
- Twitch
- Stripe
- Etsy
- Soundcloud

- Basecamp
- Kickstarter

2.4 Timeline

- First concepts and prototypes ~1993
- First release ~1995, became most popular language in Japan by 2000
- Subsequent evolution and growth outside Japan
- Ruby 3.0 released ~2020, introducing a type system for static analysis, fibers (similar to Goroutines, asyncio etc.), and completing optimizations making it ~3x faster than Ruby 2.0 (from 2013)

3 Syntax

3.1 Logic

Typical logical operators:

```
1 >> 2 < 3
2 => true
```

```
1 >> 1 == 2
2 => false
```

Comparisons are type checked:

```
1 >> 1 == "1"
2 => false
```

Triple equals can be used to check if an instance belongs to a class:

```
1 >> String === "abc"
2 => true
```

If, else, etc work as expected:

```
1 if name == "Zigor"
2   puts "#{name} is intelligent"
3 end
```

However Ruby also allows interesting variations of this, such as putting the comparisons behind the block to execute:

```
1 puts "#{name} is genius" if name == "Zigor"
```

We can also use `unless`, which is a more natural way to check for negated expressions:

```
1 p "You are a minor" unless age >= 18
```

switch statements are known as **case** statements, but don't **fallthrough** by default like in Java:

```
1 case a
2   when 1
3     spell = "one"
4   when 2
5     spell = "two"
6   when 3
7     spell = "three"
8   when 4
9     spell = "four"
10  when 5
11    spell = "five"
12  else
13    spell = nil
14 end
```

Since everything is an object, we can also use **case** statements to check if instances are of a class:

```
1 a = "Zigor"
2 case a
3   when String
4     puts "Its a string"
5   when Fixnum
6     puts "Its a number"
7 end
```

As mentioned before, Ruby is a very flexible language. The case statement for example also allows to us to check regular expressions:

```
1 case string
2   when /Ruby/
3     puts "string contains Ruby"
4   else
5     puts "string does not contain Ruby"
6 end
```

We can even use Lambdas in case statements, making long **if** ... **else** blocks unnecessary:

```
1 case num
2   when -> (n) { n % 2 == 0 }
3     puts "#{num} is even"
```

```
4 else
5   puts "#{num} is odd"
6 end
```

And the object orientation becomes very clear; we can even define our own matcher classes:

```
1 class Zigor
2   def self.==(string)
3     string.downcase == "zigor"
4   end
5 end
6
7 name = "Zigor"
8
9 case name
10 when Zigor
11   puts "Nice to meet you Zigor!!!"
12 else
13   puts "Who are you?"
14 end
```

We can also assign values from a case statement:

```
1 grade = case mark
2           when 80..100 : 'A'
3           when 60..79  : 'B'
4           when 40..59  : 'C'
5           when 0..39   : 'D'
6           else "Unable to determine grade. Try again."
7         end
```

3.2 Loops

Ruby has the **for** loop that we are all used to, but also more specialized constructs that allow for more expressive usecases:

```
1 for i in 0..10
2   p i
3 end
```

For example **upto** and **downto** methods:

```
1 10.downto 1 do |num|
2   p num
3 end
```

```
1 17.upto 23 do |i|
2   print "#{i}, "
```

```
3 end
```

Or the `times` method, which is much more readable:

```
1 7.times do
2   puts "I know something"
3 end
```

`while`, `until` and the infinite `loop` loops still exist however:

```
1 i=1
2 while i <= 10 do
3   print "#{i}, "
4   i+=1
5 end
```

```
1 i=1
2 until i > 10 do
3   print "#{i}, "
4   i+=1
5 end
```

```
1 loop do
2   puts "I Love Ruby"
3 end
```

We can also use `break`, `next` and `redo` within a loop's block:

```
1 1.upto 10 do |i|
2   break if i == 6
3   print "#{i}, "
4 end
```

```
1 10.times do |num|
2   next if num == 6
3   puts num
4 end
```

```
1 5.times do |num|
2   puts "num = #{num}"
3   puts "Do you want to redo? (y/n): "
4   option = gets.chomp
5   redo if option == 'y'
6 end
```

3.3 Arrays

Arrays in Ruby can contain multiple types and work as expected; there is no array vs collection divide:

```
1 my_array = ["Something", 123, Time.now]
```

Instead of loops you can use the `each` method to iterate:

```
1 my_array.each do |element|
2   puts element
3 end
```

We can use `<<` to add things to an array:

```
1 >> countries << "India"
2 => ["India"]
3 >> countries
4 => ["India"]
5 >> countries.size
6 => 1
7 >> countries.count
8 => 1
```

And access elements with `[0]`:

```
1 >> countries[0]
2 => "India"
```

Thanks to the `..` syntax we can also access multiple elements at once in a very simple way:

```
1 >> countries[4..9]
2 => ["China", "Niger", "Uganda", "Ireland"]
```

And use the `includes?` method (note the `?!`) to check if elements are present:

```
1 >> countries.include? "Somalia"
2 => true
```

And `delete` to delete elements:

```
1 >> countries.delete "USA"
2 => "USA"
```

If we have a nested array, using `dig` will allow us to find deeply nested elements in a simple way:

```
1 >> array = [1, 5, [7, 9, 11, ["Treasure"], "Sigma"]]
2 => [1, 5, [7, 9, 11, ["Treasure"], "Sigma"]]
3 >> array.dig(2, 3, 0)
4 => "Treasure"
```

Another very useful set of features are set operations, allowing us to modify arrays in a simple way, for example we can use the `&` operator to find elements that are in two arrays:

```
1 >> volleyball = ["Ashok", "Chavan", "Karthik", "Jesus", "Budha"]
2 => ["Ashok", "Chavan", "Karthik", "Jesus", "Budha"]
3 >> cricket = ["Budha", "Karthik", "Ragu", "Ram"]
4 => ["Budha", "Karthik", "Ragu", "Ram"]
5 >> volleyball & cricket
6 => ["Karthik", "Budha"]
```

Or `+` to merge them:

```
1 >> volleyball + cricket
2 => ["Ashok", "Chavan", "Karthik", "Jesus", "Budha", "Budha", "Karthik",
    "Ragu", "Ram"]
```

Or use `|` to merge both, but de-duplicating at the same time:

```
1 >> volleyball | cricket
2 => ["Ashok", "Chavan", "Karthik", "Jesus", "Budha", "Ragu", "Ram"]
```

Finally, we can also use `-` to remove multiple elements at once:

```
1 >> volleyball - cricket
2 => ["Ashok", "Chavan", "Jesus"]
```

For those who are familiar with MapReduce, Ruby provides all of it in the language. For example `map`:

```
1 >> array = [1, 2, 3]
2 => [1, 2, 3]
3 >> array.map{ |element| element * element }
4 => [1, 4, 9]
```

Note that this doesn't modify the array; we can use `map!` for that, which works for lots of Ruby methods:

```
1 >> array.collect!{ |element| element * element }
2 => [1, 4, 9]
3 >> array
4 => [1, 4, 9]
```

The `filter` method for example can be used in the same way (named `keep_if`, with the opposite `delete_if` also existing), and works like how you already know if from JS:

```
1 >> array = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
2 => [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
3 >> array.keep_if{ |element| element % 2 == 0 }
4 => [2, 4, 6, 8, 10]
```

3.4 Hashes

Hashes can be used to store mapped information:

```
1 mark = {}
2 mark['English'] = 50
3 mark['Math'] = 70
4 mark['Science'] = 75
```

And we can define a default value:

```
1 mark = {}
2 mark.default = 0
3 mark['English'] = 50
4 mark['Math'] = 70
5 mark['Science'] = 75
```

The hash literal `{}` also allows us to create hashes with pre-filled information:

```
1 marks = { 'English' => 50, 'Math' => 70, 'Science' => 75 }
```

To loop over hashes, we can use the `each` method again:

```
1 total = 0
2 mark.each { |key,value|
3   total += value
4 }
5 puts "Total marks = "+total.to_s
```

A very interesting feature to use in combination with hashes are symbols; they are much more efficient than strings as they are global and thus use less memory:

```
1 mark = {}
2 mark[:English] = 50
3 mark[:Math] = 70
4 mark[:Science] = 75
```

We can check this by getting their `object_id` (a kind of pointer):

```
1 c = "able was i ere i saw elba"
2 d = "able was i ere i saw elba"
3 >> c.object_id
4 => 21472860
5 >> .object_id
6 => 1441620
```

```
1 e = :some_symbol
2 f = :some_symbol
3 >> e.object_id
4 => 1097628
```

```
5 >> f.object_id
6 => 1097628
```

Just like accessing hash values is similar for arrays and hashes, we can use the same MapReduce functions on hashes:

```
1 >> hash = {a: 1, b: 2, c: 3}
2 => {:a=>1, :b=>2, :c=>3}
3 >> hash.transform_values{ |value| value * value }
4 => {:a=>1, :b=>4, :c=>9}
```

3.5 Ranges

Ranges are a cool concept in Ruby that we've used before. We can use them with the `..` notation:

```
1 >> (1..5).each {|a| print "#{a}, " }
2 => 1, 2, 3, 4, 5, => 1..5
```

We can also use them on strings:

```
1 >> ("bad".."bag").each {|a| print "#{a}, " }
2 => bad, bae, baf, bag, => "bad".."bag"
```

They can be very useful in **case** statements, where you can replace lots of **or** operators with them:

```
1 grade = case mark
2   when 80..100
3     'A'
4   when 60..79
5     'B'
6   when 40..59
7     'C'
8   when 0..39
9     'D'
10  else
11    "Unable to determine grade. Try again."
12 end
```

In addition to using them in **case** statements as described before, they can also serve as conditions:

```
1 print "Enter any letter: "
2 letter = gets.chomp
3
4 puts "You have entered a lower case letter" if ('a'..'z') === letter
5 puts "You have entered a upper case letter" if ('A'..'Z') === letter
```

We can also use triple dots, which will remove the last value:

```
1 >> (1..5).to_a
2 => [1, 2, 3, 4, 5]
3 >> (1...5).to_a
4 => [1, 2, 3, 4]
```

It is also possible to define endless ranges:

```
1 print "Enter your age: "
2 age = gets.to_i
3
4 case age
5 when 0..18
6   puts "You are a kid"
7 when (19..)
8   puts "You are grown up"
9 end
```

3.6 Functions

As mentioned before, Ruby draws a lot of inspiration from functional programming languages, and functions are a primary building block in the language as a result.

We can define functions with `def` and call them without parentheses:

```
1 def print_line
2   puts '_' * 20
3 end
4
5 print_line
```

It is also possible to define default arguments unlike in Java:

```
1 def print_line length = 20
2   puts '_' * length
3 end
4
5 print_line
6 print_line 40
```

Arguments are always passed by reference:

```
1 def array_changer array
2   array << 6
3 end
4
5 some_array = [1, 2, 3, 4, 5]
6 p some_array
7 array_changer some_array
```

```
8 p some_array
9
10 => [1, 2, 3, 4, 5]
11 => [1, 2, 3, 4, 5, 6]
```

There is no need for a **return** statements as returns are implicit (but optional for control flow support):

```
1 def addition x, y
2   x + y
3 end
4
5 addition 3, 5
6
7 => 8
```

We can also define named arguments, with or without defaults:

```
1 def say_hello name: "Martin", age: 33
2   puts "Hello #{name} your age is #{age}"
3 end
4
5 say_hello name: "Joseph", age: 7
```

Arguments can also be variadic:

```
1 def some_function a, *others
2   puts a
3   others.each do |x|
4     puts x
5   end
6 end
7
8 some_function 1,2,3,4,5
```

A very neat function is to use argument forwarding to call a function with all used parameters:

```
1 def print_something string
2   puts string
3 end
4
5 def decorate(...)
6   puts "#" * 50
7   print_something(...)
8   puts "#" * 50
9 end
10
11 decorate "Hello World!"
```

We can also define a function in more concise way:

```
1 def double(num) = num * 2
```

3.7 Classes

Besides the functional influence, Ruby is also a radically object-oriented language. As a result, it makes working with objects and classes very easy:

```
1 class Square
2 end
```

Through the `attr_reader`, `attr_writer` and `attr_accessor` notation we can add instance variables to a class:

```
1 class Square
2   attr_accessor :side_length
3 end
```

They can be read and written with `.`:

```
1 s1 = Square.new # creates a new square
2 s1.side_length = 5 # sets its side length
3 puts "Side length of s1 = #{s1.side_length}" # prints the side length
```

Methods can be defined with `def`:

```
1 class Square
2   attr_accessor :side_length
3
4   def area
5     @side_length * @side_length
6   end
7
8   def perimeter
9     4 * @side_length
10  end
11 end
```

Note the use of `@` to access instance variables.

Like many object-oriented languages, Ruby supports constructors (called initializers):

```
1 class Square
2   attr_accessor :side_length
3
4   def initialize side_length = 0
5     @side_length = side_length
6   end
```

```
7
8  def area
9    @side_length * @side_length
10  end
11
12  def perimeter
13    4 * @side_length
14  end
15  end
```

Variables defined by `attr_accessor` as public; we can make them private by omitting their definition:

```
1  class Human
2    def set_name name
3      @name = name
4    end
5
6    def get_name
7      @name
8    end
9  end
```

In a similar way, we can use **private** and **protected** to change the visibility of methods:

```
1  class Human
2    attr_accessor :name, :age
3
4    def tell_about_you
5      puts "Hello I am #{@name}. I am #{@age} years old"
6    end
7
8    private def tell_a_secret
9      puts "I am not a human, I am a computer program. He! Hee!!"
10   end
11  end
```

In addition to instance variables, we can also create class variables which work similar to static variables in Java using the `@@` notation:

```
1  class Robot
2    def initialize
3      if defined?(@@robot_count)
4        @@robot_count += 1
5      else
6        @@robot_count = 1
7      end
8    end
9
10   def self.robots_created
```

```
11   @@robot_count
12   end
13 end
```

Similarly so, we can define class constants like so:

```
1 class Something
2   Const = 25
3
4   def Const
5     Const
6   end
7 end
8
9 puts Something::Const
```

While inheritance is not the primary means of reusing code in Ruby, there is support for it in the language using the < notation:

```
1 class Rectangle
2   attr_accessor :length, :width
3 end
4
5 class Square < Rectangle
6   def initialize length
7     @width = @length = length
8   end
9
10  def side_length
11    @width
12  end
13 end
```

We can overwrite methods; interestingly it is possible to change a child's signature and use the **super** method in the child:

```
1 class Square < Rectangle
2   def set_dimension side_length
3     super side_length, side_length
4   end
5 end
```

I won't go into more details on these aspects as they are mostly similar to Java; the same goes for Threads, Exception and more. One thing uniquely powerful in Ruby is reflection; for example, you can get the methods of a class as an array using `.methods`:

```
1 >> "a".methods
2 =>
3 [:unicode_normalized?,
```

```
4  :encode!,
5  :unicode_normalize,
6  :ascii_only?,
7  :unicode_normalize!,
8  :to_r,
9  :encode,
10 :to_c,
11 :include?,
12 :%,
13 :*,
14 :+,
15 :unpack,
16 # ...
17 ]
```

We can also get private methods using `.private_methods`, instance variables using `.instance_variables` etc.

Another feature fairly unique to Ruby is method aliasing:

```
1 class Something
2   def make_noise
3     puts "AAAAAAAAAAAAAAAAHHHHHHHHHHHHHHHHHH"
4   end
5
6   alias :shout :make_noise
7 end
8
9 Something.new.shout
```

This makes it very easy to define multiple method names for things that are frequently interchanged, such as `.delete` and `.remove`, or `.filter` and `.keep_if`.

Due to Ruby's dynamic nature, we can also define classes dynamically and anonymously:

```
1 person = Class.new do
2   def say_hi
3     'Hi'
4   end
5 end.new
```

To deal with the complexities of such a dynamic language, Ruby has support for a safe navigation operator similar to Typescript:

```
1 class Robot
2   attr_accessor :name
3 end
4
5 robot = Robot.new
6 robot.name = "Zigor"
```

```
7 puts "The robots name is #{robot.name}" if robot&.name
```

3.8 Files, Modules and Mixins

We can use the `require` function to import things from files; this is very similar to how early NodeJS works:

```
1 # break_square.rb
2
3 class Square
4   attr_accessor :side_length
5
6   def perimeter
7     @side_length * 4
8   end
9 end
```

```
1 # break_main.rb
2
3 require "./break_square.rb"
4
5 s = Square.new
6 s.side_length = 5
7 puts "The squares perimeter is #{s.perimeter}"
```

However this quickly leads to problems with code organization, for example when two functions with a different purpose are named the same way. Ruby solves this issue with modules:

```
1 module Star
2   def line
3     puts '*' * 20
4   end
5 end
6
7 module Dollar
8   def line
9     puts '$' * 20
10  end
11 end
```

If we `include` `Star` and call `line`, we will print a line of stars, and if we do so with `Dollar`, calling `line` again will print dollar signs. Without including `line`, the method will be undefined.

We can also call methods and access other objects in a module using the `::` operator:

```
1 >> Dollar::line
2 => $$$$$$$$$$$$$$$$$$$$$$
```

The `include` keyword can be used to form Mixins, which will expose reusable code only to a specific class, i.e. make the `Pi` constant only accessible from a single class:

```
1 class Sphere
2   include Constants
3   attr_accessor :radius
4
5   def volume
6     (4.0/3) * Pi * radius ** 3
7   end
8 end
```

3.9 Metaprogramming

Ruby is a very flexible language, and as such it allows metaprogramming. For example, directly call a method using the `send` function by passing in the `speak` symbol:

```
1 class Person
2   attr_accessor :name
3
4   def speak
5     "Hello I am #{@name}"
6   end
7 end
8
9
10 p = Person.new
11 p.name = "Karthik"
12 puts p.send(:speak)
```

This allows for very powerful, but dangerous things, such as calling arbitrary functions by passing in the method name as a string:

```
1 class Student
2   attr_accessor :name, :math, :science, :other
3 end
4
5 s = Student.new
6 s.name = "Zigor"
7 s.math = 100
8 s.science = 100
9 s.other = 0
```

If we want to give a user access to any of the properties using `send`, we can get their input using `gets.chomp`:

```
1 print "Enter the subject who's mark you want to know: "
2 subject = gets.chomp
```

```
3 puts "The mark in #{subject} is #{s.send(subject)}"
```

We can also catch a developer calling methods that don't exist at runtime and handle that usecase explicitly by implementing a `method_missing` method:

```
1 class Something
2   def initialize
3     @name = "Jake"
4   end
5
6   def method_missing method, *args, &block
7     puts "Method: #{method} with args: #{args} does not exist"
8     block.call @name
9   end
10 end
11
12 s = Something.new
13 s.call_method "boo", 5 do |x|
14   puts x
15 end
```

As you can see, we're now able to call a method that doesn't exist, and provide the implementation ourselves:

```
1 => Method: call_method with args: ["boo", 5] does not exist
2 => Jake
```

Instead of passing in an implementation in the form of a block ourselves, we can also do other things, such as matching the incoming method name against a regular expression and then manually calling the method:

```
1 class Person
2   attr_accessor :name, :age
3
4   def initialize name, age
5     @name, @age = name, age
6   end
7
8   def method_missing method_name
9     method_name.to_s.match(/get_(\w+)/)
10    send($1)
11  end
12 end
13
14 person = Person.new "Zigor", "67893"
15 puts "#{person.get_name} is #{person.get_age} years old"
16
17 => Zigor is 67893 years old
```

It is also possible to use `define_method` to dynamically define a method at runtime:

```
1 class Person
2   def initialize name, age
3     @name, @age = name, age
4   end
5 end
6
7 Person.define_method(:get_name) do
8   @name
9 end
10
11 person = Person.new "Zigor", "67893"
12
13 >> person.get_name
14 => "Zigor"
```

We can also define class methods etc. using `define_singleton_method` or `class_eval` and `instance_eval` etc. to add arbitrary things such as `attr_accessors` to classes or even instances.

4 Usecases for Ruby

Recommended:

- Scripting
- Web Development, especially old Web 2.0-style
- MVPs in startups (see Twitter etc.)
- Applications that require excellent extensibility (see Discourse etc.)
- Applications working with highly dynamic data models
- Systems administration on UNIX (i.e. Metasploit, Chef, Puppet, Homebrew)
- “Glue code” between cloud systems (i.e. Fluentd)
- Business Intelligence apps/CRUD systems (esp. with Ruby on Rails)

Not Recommended:

- Latency-dependend/real-time applications (garbage collection)
- High throughput systems (i.e. high-RPS web services)
- Memory- or CPU-constrained systems
- Systems with static data models
- Single-binary apps/self-contained applications (use Go)
- Game or desktop application development (lack of bindings)

5 Practical Examples

5.1 dRuby

While not recommended in modern applications (see professor Kriha's "Distributed Systems" course), dRuby is an excellent example of an idiomatic Ruby way of creating servers and clients, specifically distributed objects. We can define a server like so:

```
1 require 'drb/drb'
2
3 URI = 'druby://localhost:8787'
4
5 class PersonServer
6   attr_accessor :name
7
8   def initialize(name)
9     @name = name
10  end
11
12  def local_time
13    Time.now
14  end
15 end
16
17 DRb.start_service URI, PersonServer.new('Sheepy')
18
19 puts "Listening on to URI #{URI}"
20
21 DRb.thread.join
```

And interact with the objects on the server like so:

```
1 require 'drb/drb'
2
3 URI = 'druby://localhost:8787'
4
5 DRb.start_service
6
7 puts "Connecting to URI #{URI}"
8
9 person = DRbObject.new_with_uri URI
10
11 puts "#{person.name} #{person.local_time}"
12
13 person.name = 'Noir'
14
15 puts "#{person.name} #{person.local_time}"
```

As we can see, with very little code we can get a lot of functionality.

Demo: Write such a service and expose it to the internet with `ssh -R`, then consume it

5.2 Sinatra

Aside from Ruby on Rails, Sinatra is a very neat web framework. You can define a web server in just three lines of code:

```
1 require 'sinatra'
2
3 get '/' do
4   'Hello, world!'
5 end
```

Handling POST requests and parsing data is also very simple:

```
1 before do
2   next unless request.post?
3
4   request.body.rewind
5   @request_payload = JSON.parse request.body.read
6 end
7
8 post '/' do
9   @request_payload['name']
10 end
```

By using ERB, we can render templates very easily:

```
1 require 'sinatra'
2
3 get '/' do
4   @name = params['name']
5
6   erb :index
7 end
```

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8" />
5     <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6     <meta name="viewport" content="width=device-width, initial-scale
7       =1.0" />
8     <title>ERB Learning</title>
9   </head>
10  <body>
11    <h1>Hello, <%= @name %>!</h1>
12  </body>
13 </html>
```

Demo: Add a webserver with a dRuby interface for setting the data

6 Questions